Galvo-Safe Laser DAC Firmware

Findings, fixes and measurements from a review of the ILDAWaveX16 V2 firmware against a DT50K scanner set

Oct 27, 2026 · @James Hutchinson

Abstract

The ILDAWaveX16 V2 firmware parked its scanners in a full-scale corner at every stop and boot, accepted point rates up to 200,000 per second from any client, and could leave stale samples on the output buffer after a stop. Each of these is a step input or an over-speed drive to the galvanometers, the fragile part of a laser projector, and none was visible to a user because the laser is blanked while they happen.

The review read both firmwares against the schematic, the DAC80508 data sheet, the ILDA standards, the Ether Dream and IDN protocol references and the DT50K scanner's published ratings, then measured the board with a new test tool over thirteen behaviour tests. Firmware 0.2.0 on both processors adds a parked state at centre, 10 ms blanked ramps on stop and start, a saved scanner rate limit enforced on both chips, a guard against stale packets, per-protocol drop counters and a colour-delay control.

Measured on a DT50K set: rate requests of 60k to 200k now run at the 50k limit and the client is told so; 20 of 20 stops leave the buffer empty where 4 of 5 had left 256 stale samples; IDN streams arrive at 30, 40 and 50 kpps with at most 0.2 percent loss; Ether Dream round trips fell from 14 to 8.7 ms for a 300-point command. The owner's own Modulaser run judged the test frames "dead exact on".

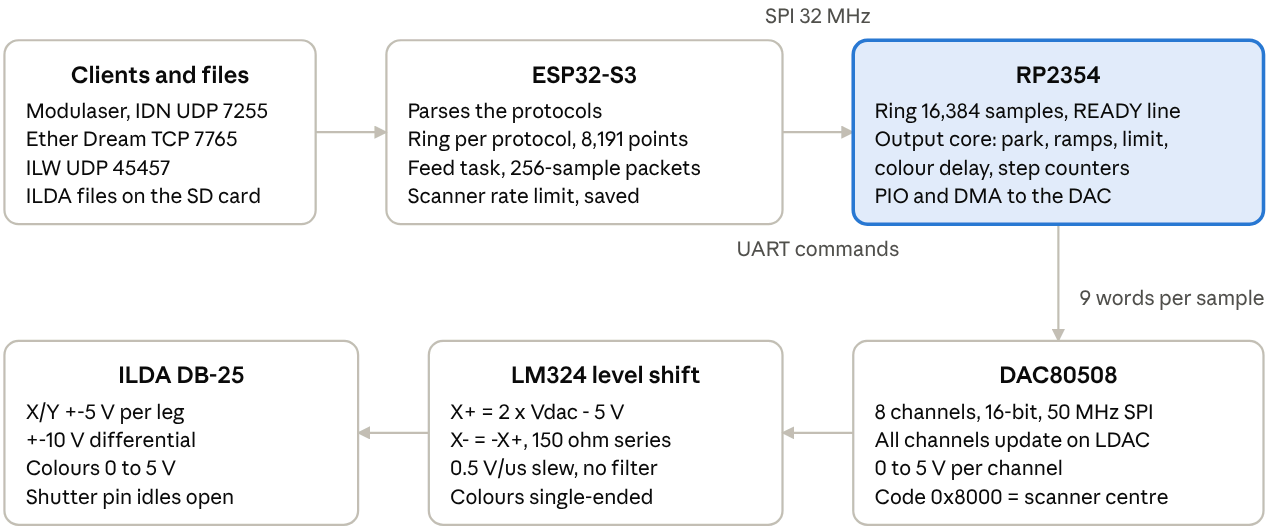
A follow-up on 2026-10-27 found why the owner saw "ghost flickers", lit lines cutting across the image: whole 256-sample packets were lost between the two chips, with nothing counted on either side. With the fix, a 20-run soak showed 0 lost packets where one run had shown 179, and the owner saw no chords over Modulaser's Ether Dream and IDN at 40 kpps.

The system

The board splits the work across two processors. The ESP32-S3 owns the network, the SD card and the web page; it parses each protocol into 8-channel, signed 16-bit samples and forwards them to the RP2354 in 256-sample SPI packets, pacing on a READY line the RP

drops when its ring is near full. The RP2354 owns timing: a second core pops samples from a 16,384-sample ring and clocks nine 24-bit words per sample into the DAC80508 through PIO and DMA, with all eight channels updating together on the LDAC trigger. A UART carries commands and a once-a-second STATUS line back.

Points cross two processors and an analog stage before they reach the scanners



data path · network or SD card to the ILDA connector, from the firmware and the REV 00 schematic

The RP is the last thing that can protect the scanners, because everything upstream has already decided what the points are; the ESP is the first place a bad request can be refused.

Why a DAC can hurt galvos

A galvanometer scanner is a coil, a magnet and a mirror on a shaft, driven by a servo amplifier that tries to make the mirror follow a +-5 V position command. It fails thermally: sustained fast, wide or square-wave drive heats the coil past its limit. A step input, where the command jumps a large fraction of full scale in one sample, is the worst case because the amplifier slews at its maximum until the mirror catches up. A DC hold, by contrast, is harmless to the galvo; it is only a burn risk for the laser.

What a DAC code means at the ILDA connector is set by the analog stage, and on this board it was the central fact of the review. The DAC80508 outputs 0 to 5 V; the LM324 level shifter computes X+ = 2 x Vdac - 5 V and a second stage inverts it for X-:

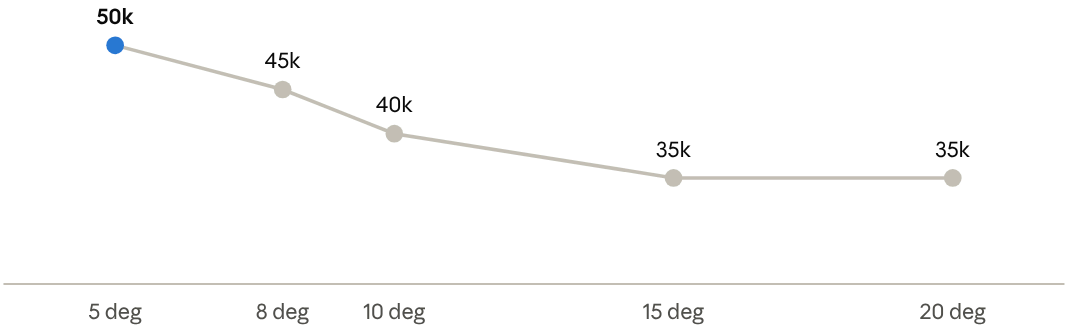
DAC code	signed sample	X+ / X-	differential	meaning
0x0000	-32768	-5 V / +5 V	-10 V	full-scale corner
0x8000	0	0 V / 0 V	0 V	scanner centre
0xFFFF	+32767	+5 V / -5 V	+10 V	opposite corner

So "all channels to zero", a natural safe state for a laser, is the worst possible state for the scanners. There is no analog filtering or slew limiting on X and Y; the LM324's 0.5 V/us slew is the only thing softening a step, and a full-scale swing takes about 20 us per stage, longer than one sample at 50 kpps. Any step limiting has to be done in firmware.

The rating of a scanner set is a speed at an angle. The DT50K is sold as 50 kpps, but the vendor's own table makes that a small-angle figure:

Rated speed falls with angle: 50 kpps at 5 degrees, 35 kpps at 15 to 20

DT50K vendor rating, thousand points per second at optical scan angle



The ILDA Test Pattern is tuned at 8 degrees; the 50 kpps headline needs 5.

Teclulu 50K galvo listing (specs match the DT50K) · 5 angles, vendor figures marked as minimums

The ILDA Test Pattern standard fixes the tuning angle at 8 degrees optical for a 30K rating and warns in capitals never to let the scanners oscillate. A comparable 50K set reviewed on laserpointerforums measured 40 kpps at 8 degrees and 17 to 19 kpps at 30 degrees, with the drivers warm at 40k. Angle and speed trade against each other, and the firmware cannot know the angle; it can only bound the speed and avoid the steps.

Method

Three research dossiers were built first, each fact cited to a page actually opened: the hardware (the REV 00 schematic rendered and read, the DAC80508 and LM324 data sheets, the ILDA ISP-DB25 standard, the DT50K's vendor data, how the Ether Dream, Helios, LaserCube and OpenLase handle underrun), the protocols (the Ether Dream protocol page and j4cDAC source, IDN-Hello and IDN-Stream, the ILDA file format, how common clients pace) and tuning practice (the ILDA Test Pattern standard, scanner-amp pots versus DAC settings, OpenLase and TouchDesigner point optimisation, Modulaser and Pangolin defaults). Both firmwares were then read end to end against those dossiers.

Every claim about behaviour was then measured on the board, never inferred from code. A test tool drives the board through the repo's own protocol clients and reads the result from the status API after a forced STATUS read of the RP, from Ether Dream status replies, from the ESP32 serial log and from the RP's USB console. It refuses to run while anything is outputting. Timing-sensitive tests run 20 times and report the pass count.

Test	What it does	Judged by
rate-request	asks the web API for 30k to 200k	applied rate
ed-begin-rate	Ether Dream Begin at 100k, blanked small circle	RP rate, rate told to the client
ed-pacing	one data command at a time, 100 to 1000 points	round trip per command
cancel-before-start	queue 1,024 points, cancel before the RP starts	RP stopped and empty, 20 runs per protocol
ed-stop-cycle	start, play 300 ms, stop	time to running, time to empty, 20 runs
ed-stop-residue	stop with the ESP send queue full	RP buffer empty, 20 runs
ed-underrun-hold	prefill 1,024 points, then silence for 1 s	still running, blanked samples, underflow flag
ed-rate-change	20k to 30k to 20k with the update command	applied rates, drops, underruns
idn-timeout	1 s of points, then silence without a close	seconds until output stops

Test	What it does	Judged by
idn-overrun	declare 30k, deliver 60k for 3 s	RP rate, ESP drop counter against client loss
idn-packet-size	64, 120 and 180 points per packet at 30k and 50k	points lost between client and RP
throughput	8 s circles at 30k, 40k, 50k, both protocols	arriving per second, underruns, drops

A second tool reports the largest lit and blanked steps between consecutive points in ILDA files, as a share of full scale. All raw tables and logs are kept with the repository.

Findings

Seventeen behaviours were found (15 to 17 on 2026-10-24 and 2026-10-27); the first four and the sixteenth put the scanners at risk directly, the rest hide problems or waste capacity. All were confirmed on the board except where noted.

#	Finding	Evidence	Status
1	STOP, CLEAR and boot wrote DAC code 0 to X and Y: a full-scale corner, held until the next START	schematic transfer function; firmware `dac_write_all_zero()`	fixed, verified
2	STOP jumped to the corner and START jumped from centre to the first point in one sample	code; LM324 slew means a step cannot complete in a sample at 50 kpps	fixed, verified
3	Any client could set up to 200 kpps; the web page offered 60k and 100k buttons	requests of 60k, 100k, 200k all applied; Begin at 100k ran at 100k	fixed, verified

#	Finding	Evidence	Status
4	A STOP with a full send queue left one 256-sample packet on the RP, played by the next START from the parked centre	4 of 5 stops left 256 samples	fixed, verified, 20 of 20
5	IDN and Ether Dream discarded samples silently when their ring was full	74,000 of 180,000 points lost with nothing in the status	counters added
6	Ether Dream clients were never told about underruns (playback flag bit 1)	flag never set during a 1 s starvation	fixed, verified
7	STATUS reads over the UART failed under load; one byte per read and a fixed 200 ms deadline while the task was starved of CPU	6 truncated replies in 23 minutes	fixed; 0 in the next hour
8	IDN loss rose with packet rate: 64-point packets lost 3.4 percent at 30k and 12 to 15 percent at 50k	packet-size sweep	0.2 percent or less after the reflash; cause not isolated
9	A repeated IDN close opened a new session	serial log	fixed
10	ILDA files on the card contain lit single-sample jumps of 56 to 72 percent of full scale	file inspector	planned: point interpolation

#	Finding	Evidence	Status
11	Each ILW CONFIG rewrites the RP's buffer watermarks in flash when they differ from ILW's constants	code	planned
12	The shutter output is never driven and idles open	schematic	planned
13	No visibility of step inputs from any source	-	step counters added, verified
14	No colour/blanking delay control	owner saw colour fringes with alignment known good	fixed, verified; test pattern redrawn to the owner's design 2026-10-27
15	Switching test patterns jumped lit to the new pattern, and Stop Stream left output started	2 lit steps of 23,000 in the owner's run; output ref left held	fixed, verified (2026-10-24)
16	Whole 256-sample packets were lost between the ESP32 and the RP when the RP's 10 ms receive timeout cut off a packet in flight, drawing lit chords across the image	179 sequence gaps in one run; 150 of 150 lit jumps at a packet's first sample	fixed, verified: 0 gaps in 20 runs (2026-10-27)
17	IDN loses about a second of points in some runs before the ESP32 sees them	4 of 20 soak runs; no ESP32 or RP counter moved	open: count IDN sequence gaps next

The step counters answered a question nobody could see before: during the test suite they read 140 to 288 for clean circles, and 20,915 to 23,737 with one to three lit big steps for others. The cause was the test client itself, which restarted its circle at point zero after

prefilling, a lit jump of 35 percent of full scale. Any client doing the same now shows up in the status.

Fixes

The design rule behind every change: the RP2354 is the last line of defence and owns anything that must be exact in time; the ESP32-S3 refuses bad requests early and reports honestly; neither retimes a client's stream.

RP2354 firmware 0.2.0

- A parked state replaces hard zero: X and Y at code 0x8000 (0 V differential, the centre), every laser channel at 0 V. Written at boot about 1 ms after the DAC reset, and on STOP, CLEAR and BUFFER.
- Soft moves: STOP ramps the scanners from their last commanded position to the centre over 10 ms with the laser blanked, then parks. START peeks at the first sample in the ring and ramps from the centre to it the same way, then plays the stream unchanged. The ramp is 16 to 1,024 samples depending on rate.
- A rate limit saved in flash (`LIMIT max_rate=`), applied to every RATE; the reply carries the applied rate. Default 30,000 pps, which any ILDA 30K scanner does at 8 degrees.
- A stream epoch: a packet whose reception straddled a STOP or CLEAR is discarded and counted, so a stale packet can never be played.
- Per-stream step counters: the largest lit X/Y step and the number of lit and blanked steps of 1/8 full scale or more.
- A colour delay line after the ring, in microseconds, applied as whole samples at the current rate, saved in flash (flashed and verified 2026-10-24).
- The output core signals idle so STOP never races a batch already being written.
- 2026-10-27: once READY is raised, the stream receive stays armed until a whole packet is in; a timeout only returns to the main loop (finding 16). New counters: packet sequence gaps, receives abandoned mid-packet, and a log line for every lit big step with its position in the packet.

ESP32-S3 firmware 0.2.0

- A scanner rate limit saved in NVS, applied to every protocol and the web page; the applied rate is reported back, Ether Dream clients see the limit as their maximum point rate, and the web page disables rate buttons above it. The ESP pushes its limit to the RP at boot and whenever the two disagree.

- STOP and CLEAR wait for the in-flight SPI packet before the RP clears its ring.
- The UART reader takes whatever is buffered in one call and extends its deadline while bytes keep arriving.
- Per-protocol drop counters; the Ether Dream underflow flag; stray IDN closes ignored.
- 2026-10-27: after each packet the sender waits for READY′s falling edge, counted in a GPIO interrupt, instead of polling for a short low it could miss and then stall 100 ms (finding 16). Its own send failures are reported in the status.

Choices that could have gone the other way. The real Ether Dream refuses a Begin above its maximum; this firmware clamps and tells the client the applied rate, because the clients in use treat a refusal as fatal and an operator mid-show is better served by a slower image than a dead one. Underrun keeps the hold-and-blank behaviour rather than the Ether Dream′s stop-and-zero, because holding position is kinder to the scanners and the flag now tells the client. The soft-start ramp costs 10 ms of latency once per stream; the first build of it re-armed itself forever and never played, which the test suite caught within a minute of flashing.

Results

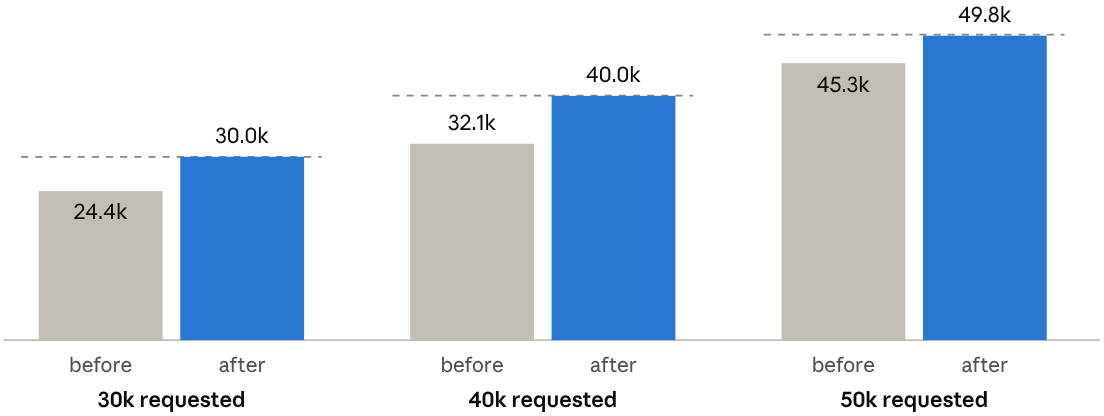
All measurements are from the same board on the same day, before and after the firmware changes, with the scanner limit set to 50,000 for the DT50K. The board itself never dropped a packet in any run.

Behaviour	Before	After
Web request for 60k, 100k, 200k pps	applied as asked	50,000
Ether Dream Begin at 100k	ran at 100k, client told 100k	ran at 50k, client told 50k
Stop with a full send queue	4 of 5 left 256 stale samples	20 of 20 empty
Cancel before START, each protocol	20 of 20 empty	20 of 20 empty
Stop cycle: Begin to running, stop to empty	57 ms median, 56 ms median	67 ms median, 48 ms median (with the 10 ms ramps)

Behaviour	Before	After
Starved stream, 1 s	held position, blanked, no flag	held position, blanked, underflow flag set
STATUS reads failing	6 in 23 minutes	0 in the following hour
IDN loss, 64-point packets at 50 kpps	11.6 percent	0.1 percent

IDN now arrives at the requested rate: 30.0k, 40.0k, 49.8k from 24.4k, 32.1k, 45.3k

Points per second reaching the RP2354 over 8 s IDN circles; dashed line = the requested rate

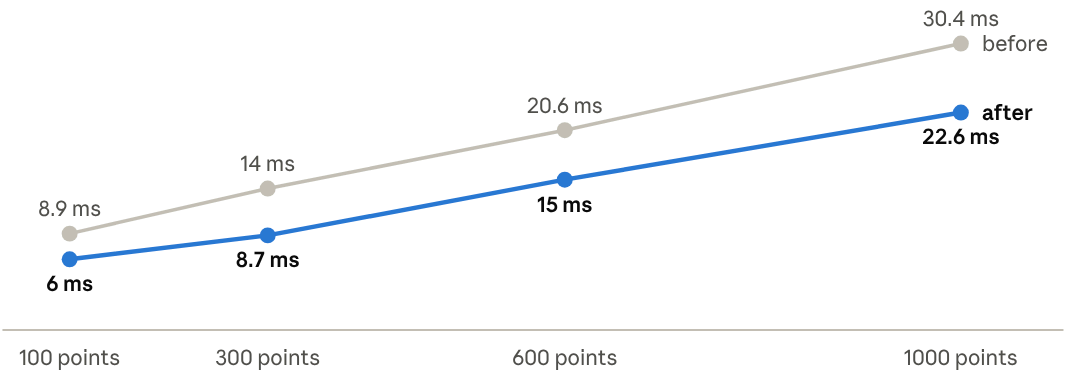


hwtest.py throughput, baseline run and the run on both 0.2.0 firmwares, 2026-10-23

The IDN gain came with the first ESP32 reflash, which also rebooted the chip; the reboot and the UART-reader change are not separated, so the cause is recorded as open.

Ether Dream round trip fell by a third: 8.7 ms for a 300-point command, from 14.0

Median milliseconds from sending one data command to its acknowledgement, one command in flight, 30 kpps



hwtest.py ed-pacing, baseline run and the run on both 0.2.0 firmwares, 2026-10-23

The round trip is about 7 ms plus 1.3 ms per kilobyte through the W5500 and lwIP; commands above the 11,520-byte TCP window show retransmit stalls in their tails. A client that waits for each acknowledgement sustains about 21 kpps at 300 points per command; one with two commands in flight reaches 30 kpps, which is where the Python test client tops out. Modulaser pipelines and did better than the test client at every rate.

The owner's run. With both 0.2.0 firmwares, Modulaser over IDN at 81 frames per second, the test frames were judged "dead exact on" by an operator with thirty years of scanner tuning. The stream was taken up to the DT50K's rated maximum and backed off:

Requested rate	Arriving per second	RP buffer, samples	Blanked underrun samples per second
28,858	28,600 to 29,000	416 to 1,024	0 to 252
40,000	39,900 to 40,200	480 to 1,568	0 to 252
45,008	44,700 to 45,200	576 to 1,824	235 to 832
50,000	49,700 to 50,250	288 to 640	1,177 to 1,729

The underruns are Modulaser's send gaps against an IDN buffer that holds only what the 3,072-sample start prebuffer left; the board had no drops. Two things the run surfaced belong to the client, not the board: rounded corners on a rotating cube (corner dwell) and a colour shift with the beams aligned (colour delay, which the owner's earlier Modulaser tuning had set against the old firmware's slow rate and lit padding).

Lost packets, 2026-10-27. One lost 256-sample packet skips about 191 degrees of the 480-point test circle, so the largest lit steps matched the circle sizes: 43,652, 23,884 and 15,922 for radii 22,000, 12,000 and 8,000. The same four tests (ed-stop-residue, ed-underrun-hold, idn-packet-size, throughput) before and after the fix:

Measure	Before	After
RP packet sequence gaps	179 in one run	0 in each of 20 runs
Receives cut mid-packet	362	0
ESP32 sender stalls of 100 ms	4 (542 with the RP half alone)	0

Measure	Before	After
Lit jumps logged	150, all at a packet's first sample	0 from the SPI link
IDN underruns per second at 50 kpps	up to 5,884	0

The owner then ran Modulaser at 40 kpps, Ether Dream at 38,757 and IDN at 39,642 points/s on average, with the laser on: no chords seen, and 0 gaps and 0 lit jumps logged.

Tuning roadmap

One rule from the research shapes where each feature lives. Per-sample transforms that do not change timing (size, invert, offset, colour delay) are safe on any stream and belong on the RP2354, after the ring, where timing is exact. Anything that adds or removes points (interpolation, corner dwell, blanking lead-in) is only for frames the ESP32 generates itself, from the SD card or its test patterns; a client's stream is never retimed, because Ether Dream and IDN clients have already optimised their points and pace by point count.

Tier	Feature	Parameter, unit, default	Where	State
1	scanner rate limit	pps, 1,000 to 200,000, default 30,000	both	done
1	parked idle at centre, blanked	-	RP	done
1	soft start and stop	us, default 10,000	RP	done; expose as a setting
1	colour / blanking delay	us, -2,000 to 2,000, default 0; practice says start near 100 to 130	RP	done; calibration pattern and one-card controls 2026-10-27
1	ILDA Test Pattern generator	12,000 or 30,000 pps, locked	ESP	planned; the file is on the card today
1	stop reason and fault latch	enum	ESP	planned

Tier	Feature	Parameter, unit, default	Where	State
2	size, invert, swap, offset	percent 10 to 100; bool; signed	RP	planned; SD player has size today
2	per-channel colour delay	us per channel	RP	planned
2	colour mapping	min and max V, gamma, TTL	RP	planned
2	stationary-beam guard	20 ms, 0.5 percent of full scale	RP	planned; the amp's scan-fail stays primary
2	speed-against-angle warning	table per scanner, DT50K above	ESP	planned
2	IDN and ILW prebuffer depth	samples, default 3,072	ESP	planned; trades latency for the underruns seen
3	max lit step, max blanked step	1 percent, 5 percent of full scale	ESP, own frames only	planned
3	corner dwell	8 repeats at 0 degrees to 0 at 180, threshold 30	ESP, own frames only	planned
3	start and end repeats, blanking lead-in	3 and 3; 100 us	ESP, own frames only	planned
4	shutter on GPIO40, colour-shift test figure, keystone, scanner-load gauge, ILW off the BUFFER command	-	-	planned

With tier 3, the grid file's 72 percent single-sample lines become drawn lines, and the RP's step counters show whether any source is still sending step inputs.

Open questions and limitations

- The DAC80508 variant is unread. The Z variant resets to code zero (a corner) and the M variant to mid-scale; the firmware now parks about 1 ms after reset, but the first millisecond after power-on is the chip's own.
- The IDN loss improvement arrived with a reboot and a code change together and was not separated. The packet-size sweep stayed clean through the rest of the session; it should be re-run after long uptime. A 20-run soak on 2026-10-27 lost about a second of IDN points in 4 runs before the ESP32 saw them (finding 17); counting IDN sequence gaps is the next step.
- The ESP re-syncs its rate limit to the RP whenever they disagree; only the agreeing case was observed after that build was flashed.
- The galvo moves themselves cannot be seen: the laser is blanked while they happen. The RP console and the status counters are the evidence; a scope on the X and Y legs would be the direct one.
- Planned instrumentation (owner): most scanner amplifiers expose a position-feedback signal that can be sampled and returned. A second Raspberry Pi Pico can sample that and any other inputs wanted, with a better external ADC where its own resolution is not enough. That would make the blanked galvo moves directly measurable. The single USB port shared by the ESP32-S3 and the RP2354 is now split into two ports by a plug-breakout splitter (upstream PR #7), so both chips are online during testing.
- Ether Dream clamps instead of refusing a rate above the limit, as the real DAC does. That is a product decision the maintainer should confirm.
- The lab has a real Ether Dream, a Helios and a Helios Pro (IDN) available for review and testing. The clamp-versus-refuse choice, underrun behaviour, pacing and IDN loss figures here can be checked side by side against them, not only against protocol pages and source. The lab also has galvo sets rated 50, 40, 30, 25, 20 and 15 kpps, but only one ILDAWaveX16 V2. The other sets run from the Ether Dream or the Helios with factory program modules. So the rate limit can be tried on this board across the whole range, one set at a time, with the reference DACs as the baseline on the others.
- The default limit of 30,000 is a proposal, and the owner recommends 15,000 as the safer default for a board like this, which leaves a margin for unknown or slower scanners. The firmware as built still defaults to 30,000; lowering it means changing the default in both firmwares (RP flash and ESP NVS). Owners of faster sets raise it once.
- Nothing here replaces the scanner amplifier's own scan-fail safeguard or the projector's interlock; the board's interlock pins are hard-wired together on the PCB and its shutter output is not driven.

References

Standards

- ILDA technical standards index
- ILDA Standard Projector, ISP99 rev 002: DB-25 pinout and signal levels
- ILDA Test Pattern, rev 002: the 12K and 30K tuning procedure at 8 degrees
- ILDA Image Data Transfer Format, rev 011
- IDN-Stream rev 002 and IDN-Hello 2022 draft
- ILDA hardware safety list and audience scanning tips

Hardware

- DAC80508 data sheet, Texas Instruments
- LM324 data sheet, Texas Instruments
- RP2350 data sheet, Raspberry Pi
- Teclulu 50 kpps galvo listing: the DT50K speed-against-angle table
- Chuyun RD6215-50K driver specifications
- Laserworld LW-50K galvo set review, laserpointerforums
- ILDA laser pinout, Pangolin

Protocols and other DACs

- Ether Dream protocol and j4cDAC firmware
- Helios DAC firmware and SDK
- OpenLase: point optimisation in libol
- laser-dac-rs, Modulaser
- Modulaser: improve laser output
- TouchDesigner Laser CHOP and EtherDream CHOP
- Pangolin legacy ILDA test pattern page

This work

- ILDAWaveX16 V2, upstream repository
- In the fork branch: `docs/galvo-safety-review.md`, `docs/research/hardware.md`, `docs/research/protocols.md`, `docs/research/ilda-tuning.md`, `tools/hwtest.py`, `tools/ild_inspect.py`, and the raw logs under `logs/`.